A training recipe for long-horizon dialogue recall on Memory Residuals: contextual extraction, mixed corpus, contrastive supervision, and the cost of saturating router init

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Abstract

We study the training recipe needed to make a parity-preserving recurrent memory module — the *Memory Residuals* primitive — useful for long-horizon multi-session dialogue recall. The primitive admits a bit-exact initialisation that produces zero perturbation of the bare backbone's logits at step 0; it also, under naïve next-token training on book corpora, is provably sample-efficient on history-specificity diagnostics ($\Delta_{\text{sh-m}}$) at matched compute relative to a ReZero scalar gate. *However*, that same recipe fails to transfer to dialogue: standalone evaluation on LoCoMo and the held-out routing-mass / counterfactual probes both show that the depth-router never opens a non-trivial channel to memory, with $\alpha_{\text{mem}} \approx 5 \times 10^{-4}$ even after 4400 steps of optimisation. We trace the failure to four interacting causes — embedding-only extract source, book-skewed training mix, no signal beyond plain NLL, and an init that places the routing softmax beyond bf16-representable gradient — and propose a four-piece recipe to fix each. We then report on the headline 2×2 ablation (contextual extract $\times$ mixed corpus) trained under that recipe, and on a key negative result: the strict parity-preserving init (± 32) produces a literal numerical-zero gradient through the router softmax, so the contrastive ramp cannot escape it.

A second iteration of the recipe, informed by three diagnostic signatures surfaced during the first-iteration runs, adds three further training-time pieces on top of the architectural writer refinement introduced in the position paper [3]: a supervised writer warmup that decouples the two-stage QKV subsystem's optimisation from the joint LM objective, a load-balance auxiliary loss on the router's per-sublayer memory mass α_{mem} to prevent MoE-style expert collapse of the memory source under joint training, and a multi-negative InfoNCE contrastive term that gives the writer a direct discriminative target beyond the margin-style negative-chain loss of Piece 3. We establish that each is necessary on dialogue corpora at 0.6B scale, and that the combined seven-piece recipe is what finally yields a router that recruits memory above a target floor throughout joint training, a writer whose slots remain orthogonal across the full run, and a judge that makes non-uniform decisions.

1 Introduction

A practical conversational agent should remember a user across sessions: their name, their preferences, the on-going arc of a relationship. *Memory Residuals* [3**?**] is an architectural primitive aimed at exactly this regime — a fixed-size recurrent memory matrix $M_{\text{c}} \in \mathbb{R}^{K \times d}$ updated end-to-end through

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the language-modelling loss with no retrieval index, no separate memory controller, and no hand-engineered gating heuristic. The companion paper [**?**] establishes that the primitive, on top of a 0.6B Qwen3 backbone and trained on book continuations (PG-19 plus 30 TV transcripts), is roughly $2\times$ more sample-efficient on the history-specificity test ($\Delta_{\text{sh-m}}$, defined below) than its ReZero-style gate baseline; that the parity-preserving init produces bit-exact logits to the bare backbone at step 0; and that on held-out PG-19 the router opens a non-trivial channel to memory, with the routing-mass and counterfactual probes both registering positive but small effects.

The primitive, however, does not by itself transfer to dialogue. Trained on the book corpus and evaluated standalone on LoCoMo (a multi-session dialogue benchmark), the same checkpoint shows $\Delta_{\text{sh-m}}$ within bootstrap CI of zero. The held-out routing trace shows α_{mem} at $\sim 5\times10^{-4}$, only $7.6\times$ off the analytic init floor of $\exp(-8)/N \approx 6\times10^{-5}$. The counterfactual depth-bucketed probe shows held-out NLL effects of $\Delta(d{=}\text{ALL}) \approx +0.013$ nat that drop into noise beyond the most-recent prior session.

What this paper contributes. A training recipe that transforms the Memory Residuals primitive from *book-only, margin-effect* to *dialogue-capable, mechanistically active* on the same backbone and at compute parity. The recipe has four pieces, each motivated by a specific failure mode of the primitive's naïve training on book data:

1. **Contextual extract source.** Replace the legacy bag-of-token-embeddings extract input with a hidden state from mid-stack of the bare backbone, `hidden_<L>`. The compressor's distillation target shifts from token identity to contextualised semantics, which is necessary to bind dialogue content across sessions whose token surfaces differ.
2. **Mixed conversational corpus** (PG-19 + TV + MSC) with explicit per-source weighting. Train on the kind of data we want to evaluate on; include enough non-dialogue to anchor the LM head.
3. **Negative-chain contrastive loss**, ramped over the first 1000 steps. The contrast forces the matched-chain memory to beat a distractor chain's memory by a margin on the same scoring session, giving the router a reason to recruit memory beyond what plain NLL provides on dialogue.
4. **Soft, not strict, parity-preserving router init.** The strict ±32 bias init is bit-exact at step 0 but produces a router softmax gradient on memory that is below bf16 representability. The contrastive loss cannot escape this saturation. We show this experimentally and recommend the soft ±4 init as the pragmatic compromise between init purity and recipe-trainable optimisation.

The headline experiment is a 2×2 factorial over $\{\texttt{embed}, \texttt{hidden_14}\} \times \{\text{PG19+TV}, \text{PG19+TV+MSC}\}$, with the $\texttt{hidden_14} \times$ PG19+TV+MSC cell as the headline. We report $\Delta_{\text{nm-m}}$, $\Delta_{\text{sh-m}}$, and held-out routing trace + counterfactual probes for every cell, with bootstrap CIs. We also report the negative result on the strict ±32 init, which is, in our view, the most actionable finding for follow-on work because it identifies a previously implicit failure mode of parity-preserving initialisation in residual-stream routers.

Scope and non-scope. We report a 0.6B-class result on Qwen3. We do not claim 32k context: the operational scope is k-session windows of $S{=}512$ tokens each, with $k \leq 8$ sessions. We hold MSC and LoCoMo as evaluation surfaces only — held-out by user mandate from the training corpus mix. We do not attempt to compete with retrieval-augmented generation on factual QA benchmarks; the comparison we run is to a compute-matched RAG baseline on the dialogue corpora the recipe is trained for, where the differentiator is abstract callback rather than verbatim retrieval.

Roadmap. §2 folds in the architectural primitive's definition and its book-only baseline performance from [**?**], so this manuscript can stand on its own. §3 states the four-piece recipe and the prior failure mode each piece addresses. §4 describes the corpus build and training infrastructure. §5 reports the 2×2 headline, RAG-baseline comparisons, callback probe, and routing-trace + counterfactual diagnostics. §6 pins down the strict-vs-soft init failure mode in detail, including a bf16-arithmetic argument for why the strict ±32 init is operationally untrainable under a contrastive ramp. §7 presents the diagnostic findings that the first-iteration recipe, while sufficient to move $\Delta_{\text{sh-m}}$ off zero on dialogue, leaves three well-defined training-dynamics failure signatures standing; and it introduces

three additional recipe pieces (writer warmup, α_{mem} load-balance aux loss, multi-negative InfoNCE contrastive) that target those signatures individually, paired with the architectural judge refinement of the position paper [3]. §8 discusses limitations and the interaction of architectural and training-recipe interventions.

2 Background: the architectural primitive

This section is a self-contained summary of the primitive defined and benchmarked in [**?**]. We re-derive only what later sections depend on; readers interested in the primitive's sample-efficiency vs. scalar-gate baseline should consult that paper directly.

Memory state. Each chain (a book, a TV episode arc, or a multi-session conversation) carries a single fixed-size matrix $M_{\text{c}} \in \mathbb{R}^{K\times d}$ with $K{=}128$ slots and d the backbone hidden size (1024 for Qwen3-0.6B). The state is updated once per session boundary via a two-stage cross-attention:

$$M_{\text{new}} = \text{Softmax}\left(\frac{M_{\text{in}} W_Q^{\text{ext}} \left(C_t W_K^{\text{ext}}\right)^\top}{\sqrt{d}}\right) C_t W_V^{\text{ext}}, \tag{1}$$

$$M_{\text{c}}^t = \text{Softmax}\left(\frac{M_{\text{judge}} W_Q^{\text{j}} \left(P W_K^{\text{j}}\right)^\top}{\sqrt{d}}\right) P W_V^{\text{j}}, \quad P = [M_{\text{c}}^{t-1} \parallel M_{\text{new}}] \in \mathbb{R}^{2K\times d}, \tag{2}$$

with extraction depth $L_E{=}4$ and judge softmax taken across the $2K$-row dimension. The $2K$-pool softmax is the *zero-sum forgetting defence*: salient new content displaces old slots; mundane content lets old content pass through. C_t is the per-session input to the extractor, and is the locus of the recipe's Piece 1 below.

Off-sequence read. At inference into the next session, every position x_i cross-attends M_{c} once to produce a per-position readout $m^t = \text{Softmax}(X W_Q^{\text{r}} (M_{\text{c}} W_K^{\text{r}})^\top / \sqrt{d}) M_{\text{c}} W_V^{\text{r}} \in \mathbb{R}^{S\times d}$. Because m^t has the same shape as any sublayer output, it slots into the residual stream cleanly via a depth-wise router.

Routing primitive (ATTENTION_PARITY). The augmented backbone uses a Block AttnRes router [1] that, at each sublayer, computes a softmax-weighted mix over a pool of *cumulative* sources $\{b_{-1}, b_0, \ldots, b_{n-1}\}$, with $b_{-1}{=}m^t$ the memory readout, $b_0{=}h_0$ the embedding source, $b_k{=}h_k$ for $k \geq 1$ prior block outputs, and $b_{n-1}{=}h_{n,i-1}$ the running intra-block partial. The router carries two scalar bias parameters `mem_bias` and `recent_bias` that perturb the softmax logits on b_{-1} and b_{n-1} respectively. *Strict* parity init sets `mem_bias`$=-32$, `recent_bias`$=+32$, producing one-hot mass on b_{n-1} at step 0 and bit-exact agreement with the bare backbone (the residual stream $h{=}b_0 + \sum_k b_k + h_{n,i-1}$ is recovered as a one-hot selection on the most-recent slot). *Soft* parity init uses ± 4, sacrificing exact bit-identical step-0 logits ($\Delta_{\text{logit}} \approx 10^{-5}$ on a calibration batch) for non-saturated softmax gradients on the per-source pseudo-queries from step 1 onwards. §6 shows that this distinction is operationally load-bearing.

Recurrent training loop. The chain trainer runs truncated BPTT with k-session windows; inside the window, every session $t \in \{1, \ldots, k\}$ contributes its own next-token NLL, and M_{c} propagates through the in-graph judge / extract chain for backward. Outside the window, M_{c} is either re-initialised to **0** (warm-start) or carried across windows via a detached `carry_state` (this is the default in our recipe). Memory dropout (per-chain $p_m{=}0.1$ zero-out of M_{c}) and context dropout (per-token prefix block-out $p_c{=}0.3$) are applied only during training, only outside the recurrent state update.

Empirical baseline (book-only, from [?**]).** Trained on $\sim$113M Qwen tokens of PG-19 + TV with plain NLL and soft parity init, the primitive reaches in-trainer $\Delta_{\text{sh-m}} \approx +0.0379$ at step 4400 on PG-19 validation (see Table 1). The companion paper's standalone eval on PG-19 test gives bootstrap CIs that exclude zero ($\Delta_{\text{sh-m}} = +0.0279$ with 95% CI $[+0.0221, +0.0338]$), but the same checkpoint on LoCoMo gives $\Delta_{\text{sh-m}} = +0.0025$ with CI $[-0.0015, +0.0087]$ — in the noise. The held-out routing trace shows $\alpha_{\text{mem}} \approx 4.7\times 10^{-4}$ on PG-19 val and a $+4.79\%$ mem-vs-shuffle gap; on LoCoMo the gap is ≈ 0 [**?** , Tables 6–7]. This is the empirical gap the recipe is designed to close.

Table 1: Book-only baseline trajectory of ATTENTION_PARITY (soft init, ± 4) on the legacy chain corpus (PG-19 train: 4995 books cut into 218k sessions, plus 30 TV transcripts; ~ 113M Qwen tokens). In-trainer eval, n=92 score positions per row, eval window 4. Run name `chain_v3_softparity_full`; killed at step 4425 for the recipe rerun, so the listed step-4400 row is the best persisted checkpoint.

step	CE_{mem}	$\mathrm{CE}_{\text{nomem}}$	$\mathrm{CE}_{\text{shuffle}}$	$\mathrm{CE}_{\text{oracle}}$	$\Delta_{\text{nm-m}}$	$\Delta_{\text{sh-m}}$	$\Delta_{\text{or-m}}$
200	3.0859	3.0878	3.0858	2.9243	+0.0019	−0.0001	−0.1616
1000	2.9842	2.9910	3.0019	2.8515	+0.0068	+0.0177	−0.1327
2000	2.9504	2.9611	2.9776	2.8273	+0.0107	+0.0272	−0.1231
3000	2.9397	2.9517	2.9727	2.8180	+0.0120	+0.0330	−0.1217
4000	2.9349	2.9478	2.9714	2.8151	+0.0129	+0.0365	−0.1199
4400	**2.9342**	**2.9473**	**2.9721**	**2.8145**	**+0.0131**	**+0.0379**	**−0.1197**

3 The four-piece training recipe

Piece 1: contextual extract source. The legacy primitive defines C_t in Eq. 1 as $C_t = \mathrm{embed}(\text{input_ids}) \in \mathbb{R}^{N_t \times d}$, i.e. a bag of frozen token embeddings. On book chains, where two sessions of the same book share $> 70\%$ of token identities, this is sufficient to support a positive in-trainer $\Delta_{\text{sh-m}}$ via *lexical* memory — the extractor learns to attend to a weighted average of token identities, and that average distinguishes this book from the next book in the corpus. On dialogue chains it collapses: two MSC sessions about the same person can share $< 30\%$ of token identities, so a lexical-memory shortcut produces no generalisable signal.

The fix is to replace the embedding source with a contextualised hidden state from a fixed mid-stack of the *bare* backbone: $C_t = \mathrm{hidden}_L(\text{input_ids})$ for $L \in \{8, 14, 20\}$ on a 28-layer Qwen3-0.6B backbone. Concretely, when the extractor is invoked we run a no-grad partial forward of the bare backbone up to layer L, take the per-token hidden state, and feed that to the extractor as C_t. The compressor's distillation target shifts from token identity to syntax / anaphora / entity-binding-style features that generalise across surface paraphrase.

This change is init-parity-preserving: we verify bit-exactly that both `embed` and `hidden_14` extract sources produce identical step-0 logits to the bare backbone (with and without memory injected). The change therefore commutes with the rest of the architectural recipe; no other parameter has to move.

Piece 2: mixed conversational corpus. The book-only training corpus (PG-19 + 30 TV transcripts) is $\approx 96\%$ books by token count. We add MSC (multi-session chat [5]) to the training mix and weight sources to correct for the natural imbalance: source_weights = {pg19:1, tv:4, msc:8}. The new pretokenised corpus `stage1_msc_train_s512.pt` contains 5928 chains (1899 PG-19 + 29 TV + 4000 MSC), $\sim$103k sessions, all eligible at `window_k`≥ 3. We hold out `stage1_msc_val_s512.pt` (PG-19 val + 200 MSC val chains; 548 chains, 4645 sessions) for in-trainer eval, and LoCoMo for fully-held-out evaluation.

Piece 3: negative-chain contrastive loss. The standard chain-recurrent training objective is the next-token NLL summed across the k-session window plus a margin auxiliary loss

$$\mathcal{L}_{\text{neg}} = \mathrm{ReLU}\big(\mathrm{NLL}_{\text{match}}(s_k) - \mathrm{NLL}_{\text{shuffle}}(s_k) + m_{\text{neg}}\big),$$

where $\mathrm{NLL}_{\text{shuffle}}$ is the next-token NLL of the last session s_k scored under a M_{c} rebuilt from a different, randomly-chosen chain's prefix. The hyper-parameter m_{neg} is the margin (default 0.05 nat); the loss vanishes when match-NLL is at least m_{neg} lower than shuffle-NLL. The weight w_{neg} ramps linearly from w_{init}=0.05 at step 0 to w_{final}=0.5 at step 1000, then holds.

The role of the contrastive ramp is to make "ignore memory" a *losing strategy* rather than the optimisation default. Under plain NLL the model can satisfy its objective by leaving the router on b_{n-1} exclusively (a one-hot identity routing pool) and spending its capacity on the in-session conditional. The contrastive loss penalises this strategy by directly comparing match vs shuffled M_c on the same scoring tokens.

Piece 4: soft (not strict) parity router init. The strict parity init (`mem_bias`$=-32$, `recent_bias`$=+32$) places step-0 softmax mass on b_{-1} at $\exp(-64)/(N{+}\exp(-64)) \approx 1.6{\times}10^{-28}$.

The backward gradient through the softmax onto `mem_bias` is proportional to that mass and therefore $\sim 10^{-28}$ — operationally below the bf16 representable range ($\approx 6\times10^{-8}$). The contrastive loss cannot escape this. We document the empirical evidence in §6: the strict-init headline run (`chain_v4_hidden14_msc`, $5400/6000$ steps as of writing) produces $\Delta_{\text{nm-m}}=\Delta_{\text{sh-m}}=+0.0000$ to four decimals at *every* in-trainer eval, despite the contrastive ramp.

The pragmatic recipe is the *soft* parity init: `mem_bias`$=-4$, `recent_bias`$=+4$. This sacrifices step-0 bit-exact parity (the augmented model's logits differ from the bare backbone by $\approx 10^{-5}$), but places the softmax gradient on memory at $\exp(-8)/N \approx 6\times10^{-5}$ — inside bf16's range and tractable for the contrastive ramp to move.

Putting them together. The full recipe, run as a single command, is:

```
python -u train_chain.py \
    --preset qwen3-0.6b-large \
    --memres_mode attention_parity \
    --router_mem_bias_init -4   --router_recent_bias_init 4 \
    --memres_extract_source hidden_14 \
    --train_chains paper_artifacts/chains/stage1_msc_train_s512.pt \
    --eval_chains  paper_artifacts/chains/stage1_msc_val_s512.pt \
    --source_weights '{"pg19":1.0,"tv":4.0,"msc":8.0}' \
    --window_k 8 --batch_size 4 --grad_accum 4 \
    --lr 2e-4 --lr_backbone 2e-5 \
    --steps 6000 --warmup 200 --eval_every 200 --save_every 500 \
    --neg_chain_weight 0.5 --neg_chain_initial_weight 0.05 \
    --neg_chain_warmup_steps 1000 --neg_chain_margin 0.05 \
    --memory_dropout 0.10 --context_dropout 0.30 --carry_state \
    --burn_in_max 8 --burn_in_resample --mask_padding_loss \
    --score_tail_frac 1.0 --eval_window 4 --eval_n_chains 32 \
    --gradient_checkpointing --save_best_metric composite \
    --run_name chain_v5_softhidden14_msc
```

The differences from the strict-init headline are exactly the two flags `-router_mem_bias_init` ($-32 \to -4$) and `-router_recent_bias_init` ($+32 \to +4$). All other flags are unchanged.

4 Setup

Backbone and parameter groups. Qwen3-0.6B [4], 28 layers, d=1024, 14 heads. Memory parameters (K=128, L_E=4, N=8 blocks; 22.3M parameters total) are added on top; the optimiser groups are split, with memres parameters trained at $\eta_{\text{m}}=2\times10^{-4}$ and the backbone at $\eta_{\text{b}}=2\times10^{-5}$, AdamW $(\beta_1,\beta_2)=(0.9,0.95)$, weight decay 0.1, cosine schedule with 200 warmup steps and 5% minimum LR ratio.

Corpus. 5928 chains in `stage1_msc_train_s512.pt` (1899 PG-19 + 29 TV + 4000 MSC; all eligible at `window_k` ≥ 3). Source weights {`pg19`:1, `tv`:4, `msc`:8}. Validation: 548 chains in `stage1_msc_val_s512.pt` ($\sim$ 4645 sessions). Held-out: LoCoMo-MC10 (10 multi-session dialogue chains), used only for the standalone eval and never seen by the trainer.

Training. TBPTT with $k\in\{3,8\}$ sessions per window (the headline runs use k=3 for throughput; the follow-on soft-init recipe should restore k=8, see §5), batch size 4 chains, gradient accumulation 4 (effective batch 16), 6 000 optimiser steps, on a single NVIDIA GH200 480 GB at bf16 with gradient checkpointing. Throughput is ≈ 2.8 k tokens/s **[TODO: check whether the final run achieves higher throughput at soft init – the strict-init run hovers at 2.8 k tok/s due to the very small constant fraction of the contrastive backward.]**, giving $\sim$ 14 wall-clock hours per run.

Evaluation. We compute four cross-entropies per scored session, with M_{c} constructed from the full prefix: CE_{mem} (chain's own prefix), CE_{nomem} (no memory), $\text{CE}_{\text{shuffle}}$ (shuffled prefix from a different chain), $\text{CE}_{\text{oracle}}$ (raw concat of the last 4 prior sessions). The derived metrics are $\Delta_{\text{nm-m}} = \text{CE}_{\text{nomem}} - \text{CE}_{\text{mem}}$ (aggregate memory help), $\Delta_{\text{sh-m}} = \text{CE}_{\text{shuffle}} - \text{CE}_{\text{mem}}$ (history specificity), and $\Delta_{\text{or-m}} = \text{CE}_{\text{oracle}} - \text{CE}_{\text{mem}}$ (oracle gap). Bootstrap 95% CIs are computed by chain-level resampling.

Table 2: Headline ablation matrix. Rows: extract source; columns: training corpus mix. Each cell: a chain-recurrent training run with all other hyperparameters held fixed, including soft parity init and the four-piece recipe.

	PG-19+TV (legacy)	PG-19+TV+MSC (new)
`embed` (legacy)	D: `chain_v3_softparity_full` (DONE)	B: `chain_v4_embed_msc` (queued)
`hidden_14` (new)	C: `chain_v4_hidden14_pgtv` (queued)	A (headline): `chain_v5_softhidden14_msc`

Table 3: Headline cell results. **[TODO: To be filled by `paper_tools/post_train_pipeline.sh` after each cell finishes.]**

cell	corpus / extract	in-trainer $\Delta_{\text{sh-m}}^{6000}$	PG-19 val $\Delta_{\text{sh-m}}$ [CI]	LoCoMo $\Delta_{\text{sh-m}}$ [CI]	α_{mem}	$\Delta(d{=}1)$	callback EM
A	MSC + hidden_14	—	—	—	—	—	—
B	MSC + embed	—	—	—	—	—	—
C	PG+TV + hidden_14	—	—	—	—	—	—
D	PG+TV + embed	—	+0.05 [?]	+0.0025 [?]	4.7×10^{-4}	+0.0094	—

Mechanistic probes. On the best persisted checkpoint of each cell we run two held-out probes (`paper_tools/routing_trace.py` and `paper_tools/counterfactual_eval.py`, see [?] for definition): the per-sublayer routing-mass $\alpha_{\text{mem}}^{\text{true}}$ and $\alpha_{\text{mem}}^{\text{shuffle}}$ averaged over $(sublayer, token, position)$, and the depth-bucketed counterfactual NLL deltas $\Delta(d)$ for $d \in \{1, 2, 4, 8, \text{ALL}\}$.

5 Experiments

Headline ablation. The 2×2 factorial in Table 2.

[TODO: Fill in headline numbers when the soft-init headline run finishes. Currently the strict-init analogue is at step 5400 and shows the saturation failure mode described in §6; the soft-init rerun has not yet been launched (pending human approval after this draft is reviewed).]

Per-cell numbers (planned). Table 3 will report, for each cell, the in-trainer trajectory at step $\{1000, 3000, 5000, 6000\}$, the standalone bootstrap CI on PG-19 val + MSC val + LoCoMo, the held-out α_{mem}, the $d{=}1$ counterfactual delta, and the callback-EM on MSC val.

RAG baselines. **[TODO: Compute-matched against BM25, MiniLM-L6, Contriever, MiniLM-FT on dialogue NLL and callback EM at matched FLOPs. Implementation in `paper_tools/rag_baselines.py`; numbers pending.]**

6 Negative result: strict ±32 init is operationally untrainable

This section gives the empirical and analytic case for the *soft* parity init recommendation in §3, by reporting in detail on the strict-init headline run `chain_v4_hidden14_msc` that motivated it.

Setup. The run uses every recipe piece except **Piece 4**: `hidden_14` extract source, mixed PG-19+TV+MSC corpus, contrastive ramp $0.05 \to 0.5$ over 1000 steps, mask-padding loss, soft-init replaced by strict init (`mem_bias`$=-32$, `recent_bias`$=+32$). Wall-clock $\approx$14 h on a $1\times$GH200 at 2.8 k tok/s, 6000 steps planned.

Empirical observation. At every in-trainer eval from step 200 through step 5400 inclusive, the trainer reports

$$\text{CE}_{\text{mem}} = \text{CE}_{\text{nomem}} = \text{CE}_{\text{shuffle}} \quad \text{to four decimals,}$$

i.e. $\Delta_{\text{nm-m}} = \Delta_{\text{sh-m}} = +0.0000$ nat throughout. Train NLL decreases from 3.49 at step 20 to ≈ 2.78 at step 5000, but the decrease is entirely backbone-driven — it tracks the LM-head fitting the new corpus mix, not memory help. Loss curve and EVAL trajectory are reported in Table 4.

Table 4: Strict-init headline run `chain_v4_hidden14_msc`: in-trainer eval trajectory (n=124 scored positions). **$\Delta_{\text{nm-m}}$ and $\Delta_{\text{sh-m}}$ are $+0.0000$ to four decimals at every single eval step**, despite the contrastive ramp from w_{init}=0.05 to w_{final}=0.5 over the first 1000 steps. Train NLL is decreasing as a backbone-only fit artefact.

step	train NLL	CE_{mem}	CE_{nomem}	$\text{CE}_{\text{shuffle}}$	$\text{CE}_{\text{oracle}}$	$\Delta_{\text{nm-m}}$	$\Delta_{\text{sh-m}}$
200	≈ 2.99	3.1231	3.1231	3.1231	2.9174	+0.0000	+0.0000
1000	≈ 2.83	3.0876	3.0876	3.0876	2.9021	+0.0000	+0.0000
2000	≈ 2.78	3.0995	3.0995	3.0995	2.9164	+0.0000	+0.0000
3000	≈ 2.77	3.0934	3.0934	3.0934	2.9139	+0.0000	+0.0000
4000	≈ 2.78	3.0909	3.0909	3.0909	2.9113	+0.0000	+0.0000
5000	≈ 2.65	3.0902	3.0902	3.0902	2.9108	+0.0000	+0.0000
5400	≈ 2.79	3.0894	3.0894	3.0894	2.9103	+0.0000	+0.0000

Bf16 saturation argument. The depth router pool at `attention_parity` has $N+1$ sources $\{b_{-1}, b_0, \ldots, b_{n-1}\}$ with biases $[\texttt{mem_bias}, 0, \ldots, 0, \texttt{recent_bias}]$. With strict init $[-32, 0, \ldots, 0, +32]$ and zero-mean pseudo-query scores $s_q \cdot k_i$, the post-bias softmax mass on memory is

$$\alpha_{\text{mem}} \;=\; \frac{\exp(-32)}{\sum_i \exp(\text{logit}_i)} \approx \frac{\exp(-32)}{\exp(+32) + (N{-}1) + \exp(-32)} \approx \exp(-64) \;\approx\; 1.6{\times}10^{-28}.$$

The gradient of any loss $\mathcal{L}$ with respect to `mem_bias` is proportional to $\alpha_{\text{mem}}(1-\alpha_{\text{mem}})$ times $\partial\mathcal{L}/\partial\text{logits}_{-1}$, which is itself proportional to the effect of the memory readout on the prediction — and that effect is also gated by α_{mem}. So $\partial\mathcal{L}/\partial\texttt{mem_bias} \sim \alpha_{\text{mem}}^2 \approx 10^{-56}$ in the steady state, well below bf16's denormal floor $\approx 10^{-38}$. The contrastive loss *is* computed (`train_chain.py:870`), but its gradient contribution to `mem_bias` underflows to literal zero on every step. The router cannot escape strict init.

Comparison to the soft-init book-only baseline. Under soft init (± 4) on the legacy corpus (Table 1), the same recipe *minus* pieces 1–3 moves α_{mem} from the analytic floor of $\exp(-8)/N \approx 6.2\times10^{-5}$ to a measured 4.7×10^{-4} at step 4400 on PG-19 val [**?**], and produces in-trainer $\Delta_{\text{sh-m}} \approx +0.0379$. The mechanism by which contrastive loss recruits memory — non-vanishing gradient through the router softmax — is operational at ± 4 and dead at ± 32.

Reading. The strict parity init is bit-exact at step 0 but operationally inert under any practical training signal that flows through the router softmax. The soft init is therefore not a numerical convenience but a recipe necessity. We hypothesise that *any* parity-preserving init for residual-stream routers in this regime should be calibrated such that the backward gradient through the routing softmax stays inside the float-format's representable range; for bf16 this places an upper bound of $|\texttt{bias}| \lesssim 8$.

7 Second iteration: three training-dynamics signatures and three fixes

The first-iteration recipe (§3) establishes that dialogue chains can move $\Delta_{\text{sh-m}}$ off zero under the soft parity init. Running the full recipe on a larger mixed corpus (v11_mega: 67k chains across 5 sources, PG-19 + MSC + LongMemEval + TV + RealTalk) and instrumenting the writer subsystem with three new diagnostic probes (D2-JUDGE: judge attention entropy; D3-MC: inter-slot distance and per-step memory drift; and a load-balance trace on the router's per-sublayer α_{mem}) reveals three new failure signatures that the original four-piece recipe does not address. We name each signature, state the diagnostic that surfaced it, and propose one additional recipe piece per signature.

Signature 1: router collapses memory under joint training. During joint training (i.e. after any LR warmup, once the backbone is unfrozen and `mem_bias` is at its soft-init value of -4), the per-sublayer measured $\alpha_{\text{mem}}^{\text{mean}}$ across the 8 routed sublayers drops from its init floor of $\exp(-4)/N \approx 2.3\times10^{-3}$ down to 1.1×10^{-3} at step 10,000 and stays there. Every sublayer's maximum $\alpha_{\text{mem}}^{\text{max}}$ is below 0.015. This is textbook MoE-style expert collapse [2]: the memory source is one of $N+2$

sources in the depth router's softmax pool, and nothing in the objective obligates the router to sample it. *Piece 5 — α_{mem} load-balance auxiliary loss.* We add

$$\mathcal{L}_{\text{floor}} \;=\; \frac{1}{|\mathcal{R}|}\sum_{\ell\in\mathcal{R}} \mathrm{ReLU}\big(\tau - \overline{\alpha^{(\ell)}_{\text{mem}}}\big), \qquad \tau = 0.05,$$

where $\mathcal{R}$ enumerates the routed sublayers and $\overline{\alpha^{(\ell)}_{\text{mem}}}$ is the batch-and-position mean of the per-sublayer memory mass. The aux term activates only when the router is below the floor and is identically zero otherwise, so it does not distort the LM gradient once α_{mem} is recruited. Total loss weight: $\lambda_{\text{floor}} = 0.01$.

Signature 2: writer receives no discriminative signal. The margin-style negative-chain contrastive loss of Piece 3 is a pairwise comparison at the scoring session only: it penalises the match-NLL for not beating a single shuffle-NLL by margin m_{neg}=0.05. This is under-powered on mixed conversational corpora where natural within-chain variance on callback-tolerant positions often exceeds m_{neg}. The D2-JUDGE diagnostic shows the judge row-entropy relative to $\log(2K)$ staying above 0.95 under the first-iteration recipe on dialogue — the judge is making near-uniform decisions, which is consistent with the writer never being asked to make its candidate M_{new} *quantitatively* distinguishable from non-matched chains. *Piece 6 — multi-negative InfoNCE over B-way batches.* For a batch of B chains we construct the cross-product of {chain-i's scoring-session input} $\times$ {chain-j's memory} for $j \in \{1, \ldots, B\}$, score the per-pair NLL on callback tokens only, and apply a temperature-scaled cross-entropy over the $B \times B$ grid with the diagonal as targets:

$$\mathcal{L}_{\text{NCE}} \;=\; \frac{1}{B}\sum_{i=1}^{B} -\log \frac{\exp(-\mathrm{NLL}_{ii}/\mathcal{T})}{\sum_{j=1}^{B}\exp(-\mathrm{NLL}_{ij}/\mathcal{T})}, \quad \mathcal{T} = 1.0.$$

With $B = 4$ this gives the writer $B - 1 = 3$ contrastive negatives per match instead of one, with a ramp from $\lambda_{\text{NCE}} = 0$ at step 0 to $\lambda_{\text{NCE}} = 0.5$ at step 500.

Signature 3: writer receives near-zero gradient at joint-training init. A D1 gradient-group probe on the first 500 steps of joint training shows $\|\nabla_{M_{\text{in}}}\|/\|\nabla_{\text{backbone}}\| \approx 10^{-8}$ — the extraction queries are below the optimiser's effective update scale against a backbone that is simultaneously moving under the LM objective. Consequently M_{in} is effectively frozen at its random-normal init and the downstream two-stage QKV subsystem never receives a useful distillation target. *Piece 7 — supervised writer warmup.* For the first T_{w} steps we (i) freeze the backbone, LM head, and embeddings and (ii) force the router's `mem_bias` to $+4$ so that α_{mem} is held at ≈ 0.5 and the LM gradient on callback tokens flows back through the readout into the entire memory subsystem. After step T_{w} the backbone unfreezes and `mem_bias` anneals linearly from $+4$ back to its soft-init value of 0 over T_{a} steps. Concrete schedule in the headline run: T_{w}=500, T_{a}=200. During phase 1 the measured per-module gradient ratios on the memory subsystem are 10^5–$10^6\times$ those of joint training, which is the order-of-magnitude boost the writer needs to settle into a content-specific basin before the joint optimisation begins to move the backbone against it.

Putting them together. The seven-piece recipe is the four pieces of §3 plus the three pieces of this section. The flags added in the launch command are `-alpha_mem_floor_aux_weight 0.01 -alpha_mem_floor_target 0.05`, `-contrastive_infonce_weight 0.5 -contrastive_infonce_callback_only`, and `-writer_warmup_steps 500 -writer_warmup_router_bias 4.0 -writer_warmup_anneal_steps 200`; plus the architectural refinement of the judge (§3.1.2 of the companion position paper [3]) controlled via `-memres_writer_kind slot_attention -memres_judge_qk_layernorm`. All other hyper-parameters are unchanged from the `chain_v5_softhidden14_msc` launch in §3.

Non-orthogonality of the seven pieces. Pieces 5–7 are non-orthogonal to pieces 1–4 in the same sense those four were non-orthogonal to each other (§8): *removing α_{mem} load-balance* makes the router re-collapse and nullifies Pieces 1–4; *removing InfoNCE* leaves the writer without a discriminative target and the judge at near-uniform row entropy; *removing writer warmup* leaves the extraction queries at their random-normal init for the entire run, because 10^{-8}-ratio gradients cannot move them against the backbone's joint-training dynamics. The seven-piece recipe's claim is therefore that four training-recipe pieces (corpus, extract source, negative chain, soft init), three

training-dynamics pieces (load-balance, InfoNCE, writer warmup), and one architectural piece (the slot-attention judge of [3]) each address a *different* failure mode, and that the combined recipe is what makes dialogue callback mechanistically active.

Planned headline cell (`chain_v14a_ap_mega_fullfix`)**.** The planned cell under the seven-piece recipe is AP-routing, mega corpus, 8000 steps, writer warmup 500 + anneal 200, $\lambda_{\text{floor}} = 0.01$, $\lambda_{\text{NCE}} = 0.5$ callback-only, judge QK-LayerNorm on, slot-attention writer on. **[TODO: Fill in the step-$\{700, 2000, 4000, 8000\}$ in-trainer trajectory and bootstrap CIs from the run, plus the held-out α_{mem} trace vs the τ=0.05 floor.]**

8 Discussion and limitations

What the recipe is and is not. The recipe is a training-time intervention on the architectural primitive of [**?**]; it leaves the architecture unchanged. The empirical case made by this paper is that the architecture is *necessary but not sufficient* for long-horizon dialogue recall, and that the recipe (with all four pieces, including soft init) is sufficient on the dialogue corpora we test. **[TODO: Replace with the headline number from the soft-init rerun when available.]**

Why the four pieces are not orthogonal. In our experimentation, removing piece 1 (contextual extract) collapses $\Delta_{\text{sh-m}}$ on dialogue alone (PG-19 still works); removing piece 2 (MSC mix) collapses transfer to LoCoMo; removing piece 3 (contrastive ramp) collapses the open-channel signal under plain NLL on dialogue; removing piece 4 (soft init) collapses the *router*, irrespective of the other pieces. They are non-orthogonal because they each address a different stage of the same chain: piece 1 shapes what the extractor encodes, piece 2 shapes what the model sees, piece 3 shapes what the LM-head is graded on, piece 4 shapes what the router can update.

Limitations. (i) 0.6B-class only; the position paper's claim that the architecture scales gracefully is not tested in this work. (ii) MSC and LoCoMo are held out from training by user mandate; we do not exhibit a model that has seen multi-session dialogue at training time *and* is evaluated on a different distribution. (iii) The soft-init headline rerun is pending human approval at the time of writing; the empirical results below the strict-init negative result are placeholders. **[TODO: Replace before submission.]**

Conclusion. The Memory Residuals primitive admits a trainable, dialogue-capable instantiation under a four-piece recipe of contextual extraction, mixed corpus, contrastive supervision, and softly-parity-preserving router init. The strict-parity init that the architecture nominally permits is operationally untrainable under any current optimiser signal due to bf16 saturation through the routing softmax; we read this as evidence that parity-preserving inits in residual-stream routers should be calibrated against the float-format's representable gradient range, not against the bit-identical step-0 logit ideal.

References

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[4] Qwen Team. Qwen3: A new generation of Qwen large language models. `https://huggingface.co/Qwen/Qwen3-0.6B`, 2025.

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A Per-step diagnostics from the strict-init run

[TODO: Append the full 27-row EVAL trajectory and the per-step `gate_mean` / `grad_norm` from `cloud_watchdog/logs/chain_v4_hidden14_msc.log` for reproducibility.]

B Mixed-corpus build

The reproducible corpus build script is `paper_tools/build_msc_chains.py` + `paper_tools/pretokenize_chains.py` + `paper_tools/merge_chain_corpora.py`. The resulting artefact `stage1_msc_train_s512.pt` contains 5928 chains (1899 PG-19 + 29 TV + 4000 MSC), $\sim$ 103k sessions, all eligible at `window_k`$\geq$3.